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E-220 STATION BATTERIES AND CHARGERS

1. SCOPE OF WORK

The work shall include the design, supply, factory testing, delivery to the site, installation, testing and commissioning of the power station batteries and their associated chargers in accordance with Part 4, Subsection 1.2, Summary of Work.

The 220 V and 24 V batteries for each Steam Turbine unit shall be furnished complete with mounting racks, all specified and/or required accessories and associated battery chargers as follows:

- (1) **One (1)** x 100% set of 220 V, station power battery consisting of series connected, vented lead acid type, storage cells with interconnection (one for operation the other for standby) for power plant.
- (2) Two (2) x 100% set of 220 V, Silicon controlled rectifier (SCR) type battery chargers, each sized 100% required capacity with interconnection (one for operation the other for standby) (100% redundancy) for power plant.
- (3) One (1)x 100% set of 24 V, station power battery consisting of series connected, vented lead acid type, storage cells for instrumentation and control (if applicable).
- (4) Two (2) x 100% set of 24 V, silicon controlled rectifier (SCR) type battery chargers, each sized 100% required capacity with interconnection (one for operation the other for standby) (100% redundancy) for instrumentation and control (if applicable).

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The Contractor shall be responsible for final sizing of the station batteries and their associated chargers in accordance with the industry standards and arrangement the Owner's criteria. The Contractor shall furnish and install the station batteries and their chargers as finally sized.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards:

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation – Indonesia)
- SPLN : Standard Perusahaan Umum Listrik Negara

2.2 Institute of Electrical and Electronics Engineers

- IEEE 450 : Recommended Practice for Maintenance, Testing, and Replacement of Vented Lead-acid Batteries for Stationary Applications
- IEEE 484 : Recommended Practice for Installation Design and Installation of Vented Lead-acid Batteries for Stationary Applications
- IEEE 485 : Recommended Practice for Sizing Lead-acid Batteries for Stationary Applications

2.3 National Electrical Manufacturers Association

- NEMA PE5 : Utility Type Battery Chargers

2.4 International Electrotechnical Commission

- IEC 60051 : Direct acting analogue electrical measuring instruments and their accessories
- IEC 60146 : Semiconductor converters
- IEC 60529 : Degrees of protection provided by enclosures (IP Code)
- IEC 60947 : Low-voltage switchgear and controlgear

3. OPERATING AND DESIGN CONDITIONS

3.1 General

The equipment shall be suitable for operation at rated output in a combined cycle power plant on a tropical seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for detailed information on meteorological, seismic and environmental conditions at the Site.

The station DC system shall be ungrounded and shall operate at 220 volts and 24 volts (if applicable). The 220 volts system shall supply DC Power to DC motors, uninterruptible power supply inverters, emergency DC lighting, control power to switchgear, relays, plant annunciator, etc. The 24 volts system shall supply DC power for instrumentation and control system for the power plant, if applicable.

The station DC systems shall be on unit basis and each system shall consist of a separate battery, an associated battery charger and separate distribution panel.

3.2 Batteries

The 220 volts batteries shall normally operate at 234 volts float charge. The maximum and minimum voltages at the battery terminals shall be 242 and 187 volts, respectively. The maximum and minimum battery voltages are based on 2.33 volts per cell for charge equalization and 1.80 volts per cell end of load cycle discharge voltage, respectively. The 24 volt batteries shall operate with the adequate float charge voltage, the maximum and minimum voltages on the requirement of the Contractor's scope of work, if applicable.

The Contractor shall verify the estimated battery loads using the actual load data and/or conservative load estimates. It shall be the Contractor's responsibility to size and arrange the station batteries adequately for the intended service and for the specific cell types he proposes to furnish. The Contractor shall submit his battery sizing criteria and calculations including the characteristic curves for the proposed cells for the Owner's approval. Calculations shall be in accordance with IEEE-485 standard.

3.3 Battery Chargers

The battery chargers shall normally float charge the batteries and shall also supply the normal DC load. The battery chargers shall equalize the batteries when needed. They shall also be capable of recharging their respective batteries in 12 hours at the end of the battery discharge load cycle while supplying the normal load of their respective batteries.

The Contractor shall verify the estimated loads using the actual load data and/or conservative load estimates. It shall be the Contractor's responsibility to size the battery chargers adequately considering the recharge requirements for his proposed battery cells. The Contractor shall submit his battery charger sizing criteria and calculations, including the recharge characteristic curves for the

proposed cells, for the Owner's approval.

The AC power supply to the battery chargers shall be 380 volts, 3-phase, 50 Hz. Refer to Part 4, Technical Specification E-001 for details.

4. DETAILED REQUIREMENTS

4.1 Batteries and Racks

4.1.1 General

The batteries shall be rated in amperes and ampere-hours on 10 hour basis to a specified end voltage and temperature. Aging factor of 25% shall be used in the calculation, and two (2) spare cells shall be installed on the rack.

The batteries shall be vented lead acid for stationary service in a power plant.

The batteries shall include the individual cells, intercell and inter-row connectors, terminal lugs, racks, cell numbers, and accessories as hereinafter Specified.

The batteries shall be supplied dry-charged with electrolyte in separate containers. Sufficient electrolyte shall be provided to permit the first filling of each cell on site and for topping up during the commissioning and through the one (1) year warrantee period.

Each battery shall have two (2) spare cells, installed and ready to be connected in parallel with two (2) end cells of the battery.

The 220 V batteries will be charged with silicon controlled rectifier type chargers. They normally be float charged at 2.25 volts per cell and will be equalized at a maximum of 2.33 volts per cell, when necessary. The float charge voltage may vary from 2.17 to 2.25 volts per cell.

The 24 V batteries will be charged with silicon controlled rectifier type chargers with the adequate float charge and equalization requirements, if applicable.

Each storage battery shall be provided with a battery rack. The racks shall be arranged in steps so that none of the cells are directly above the others. The racks shall be of all steel construction with non-aging plastic or molded rubber covers for the battery rails.

4.1.2 Ratings

Unless otherwise specified, the batteries shall be rated in accordance with the following:

- (1) The cell voltage at the end of each of the discharge periods shall not be less than 1.75 volts, including the resistance drop of one (1) intercell connector in series with the cell.
- (2) The batteries shall be rated for 3 (three) hours of operation without the battery charger.
- (3) The discharge capacity of each 220 volts battery shall be sufficient to supply the following loads during the discharge period:
 - a. Operate all switching devices (circuit breakers, disconnecting switches, etc.) within their respective operating voltage limits while supplying the normal continuous load for the full discharge period.
 - b. When the battery is at the end of its discharge period, it shall be capable of performing the tripping duty required by the worst (for battery drain) fault, the ambient temperature being the maximum stated.
 - c. The discharge capacity of the battery shall be sufficient to supply the connected loads for 3 hours with a 25% design margin specified in IEEE-485 and two (2) additional connected spare cells. Sizing shall be by the method in IEEE-485.
 - d. The float voltage of the battery shall be the optimum required to ensure the maximum possible period between trooping up with electrolyte, without the need for an equalizing charger to restore the battery to its fully charged condition. The equalizing voltage of the complete battery shall not exceed the maximum voltage rating of the equipment being supplied.
 - e. The batteries shall be designed for operation on two-wire ungrounded system.

f. Supporting calculations to justify the battery size shall be submitted for approval.

(4) The discharge capacity of 24 volts shall be sufficient to supply the loads as mention in item (3) above during the discharge period, (if applicable).

4.1.3 Accessories:

Each battery shall be furnished with the following accessories.:

- (1) One (1) thermometer, and One(1) hydrometer vent-hole type.
- (2) One (1) set of wrenches.
- (3) One (1) hand-operated hoist and One(1) hand-operated lifter.
- (4) One (1) acid-resistant face shield and pair of gloves
- (5) One (1) container for acid (adequate for one (1) year's service need).
- (6) One (1) container for distilled water (adequate for one (1) year's service need).
- (7) One (1) cell bridging connector.
- (8) One (1) funnel.
- (9) One (1) eye irrigation device and first-aid kit.

4.2 Battery Chargers

Each battery charger shall be a full wave silicon controlled rectifier type with fully automatic controls and shall conform to applicable IEC and NEMA Standards.

The cabinet shall be arranged for floor mounting, and shall have a degree of protection of IP 33 in accordance with IEC 60529.

Each battery charger shall be capable of fully charging its corresponding battery upon initial installation. It shall be capable of being connected to the fully discharged battery, without overloading and recharging it to 80% of its full capacity within a period of 24 hours, while simultaneously supplying the normal continuous load. Each battery charger shall be rated for 380 volt, 3 phase, 50 Hz system.

Each battery charger shall be capable of float charging the battery at 2.17 to 2.25 volts per cell and simultaneously supplying other loads to its full ampere.

Each battery charger shall be capable of providing an equalizing charge at 2.25 to 2.33 volts per cell.

The input transformers of the battery chargers shall have isolated secondary windings to isolate the AC supply from the DC output. The input transformers shall have four (4) equal voltage adjustment taps at $\pm 2.5\%$ and $\pm 5\%$ of the line voltage in addition to the center tap.

The 220 V DC output voltage shall be independently and continuously adjustable from 220 to 250 volts. The 24 V DC output voltage shall be independently and continuously adjustable between the adequate voltage range or required, if applicable.

Each battery charger shall have a voltage regulation of $\pm 0.5\%$ from no load to full load with $\pm 10\%$ supply voltage variation. It shall operate properly over $\pm 5\%$ supply frequency variation. It shall be provided with an automatic load limiting feature that will limit the output current to 110% of its rated load without tripping the AC or DC breaker or blowing fuses. It shall also be capable of picking up a completely discharged battery without tripping.

The voltage ripple in the charger output shall not exceed 0.1% rms without connected to the battery.

Each battery charger shall be designed to prevent the battery from discharging back through the charger in case of AC power failure or other charger malfunction.

Adequacy of the battery charger ratings determined by the Contractor shall be subject to the Owner's approval.

Each battery charger shall be equipped with the following:

- (1) Output DC voltmeter, 90 mm scale, 2.0 % accuracy at full scale.
- (2) Output DC ammeter, 90 mm scale, 2.0% accuracy at full scale.
- (3) Float voltage potentiometer.

- (4) Equalizing voltage potentiometer.
- (5) Float and equalizing switch. (The float and equalizing voltage potentiometers shall be independent of each other).
- (6) AC power failure relay with contacts to close on loss of AC power.
- (7) Ground detection relay with test switch.
- (8) DC low voltage alarm relay with contacts to close on low output voltage or loss of charger output. The alarm point shall be adjustable from 170 to 210 volts for the 220 V battery. The alarm point shall be adjustable between the adequate voltage range as required for the 24 V battery, if applicable.
- (9) Input AC thermal magnetic circuit breaker, * kA, 3 pole, with DC shunt trip coil for 220 V battery charger, and 24 V charger, if applicable. (* Item to be finalized by the Contractor and reviewed by the Owner).
- (10) DC high-voltage alarm relay adjustable from 220 volts to 280 volt for the 220 volt battery charger. DC high-voltage alarm relay adjustable from between the adequate voltage range or required for the 24 V battery charger, if applicable. The relay shall have a 30 second fixed time delay. The contacts shall close on sustained high voltage and trip the AC input circuit breaker with a shunt trip coil.
- (11) AC power “ON” pilot light.
- (12) Equalizing charge pilot light.
- (13) 0-72 hour manual-reset equalizing charge timer.

The battery charger nameplate shall contain:

- (1) Manufacturer’s name, model, type, and serial number.
- (2) AC voltage and frequency Number of phases.
- (3) AC aand DC mpere rating.
- (4) DC voltage rating
- (5) Ripple voltage in mV when connected to fully charged battery.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 Batteries

The batteries shall be given type and routine tests to prove that they will meet the specified 10 hour discharge rate.

5.2 Battery Chargers

Converters shall be given type and routine tests in accordance with IEC 60146, or equivalent. The input and output insulation levels shall be capable of withstanding a test voltage of 1,000 volts plus two (2) times the line voltage between current-carrying and non-current-carrying parts for one (1) minute on the power circuits.

The charger shall be capable of passing the following Owner's acceptance tests:

- (1) Four-hour heat run at current limiting value.
- (2) Voltage regulation check from 0 to 100% rated output with $\pm 10\%$ AC supply voltage variation.
- (3) Voltage ripple measurement with an oscilloscope while the charger is connected to a fully-charged battery and carrying full load.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for batteries and battery chargers shall conform to the requirements in Part 4, Subsection 2.7.

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